Prashant

cs24btech11050@iith.ac.in | prashantsinghaniya01@gmail.com | +91 7017130954 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.97
PM SHRI SCHOOL J N V BAGALUR BANGALORE URBAN KK	Class XII (CBSE)	2024	90.60%
JAWAHAR NAVODAYA VIDYALAYA BHOGAON MAINPURI UP	Class X (CBSE)	2022	89.00%

SKILLS

Languages: C++, C, Java, Python, JavaScript, Assembly (RISC-V), Bash, HTML/CSS

Technical Tools: Git, Github, Markdown, LaTeX, NumPy, Pandas, Matplotlib, Docker, Fusion 360, Linux

RELEVANT COURSES

Computer Science: Software Development Fundamentals, Data Structures, Algorithms, Discrete Mathematics, Theory of

Computation, Operating Systems, Database Management Systems, Computer Architecture, Compilers I

Mathematical Foundations: Calculus I & II, Linear Algebra, Probability and Random Variables, Introduction to Statistics, Transform Techniques

PROJECTS

Register-Level Memory Optimization for Matrix Multiplication & FFT | *RISC-V Assembly, gem5* | {GitHub}

- Engineered register-tiling optimization in RISC-V Assembly, reducing load/store instructions and achieving a **3.65×** speedup for matrix multiplication using gem5 profiling and cycle-level performance measurements.
- Designed a dynamic register allocation strategy for matrix multiplication and FFT, reducing instruction count by 73% while maintaining consistent performance across different inputs and latency configurations.
- Benchmarked an optimized RISC-V FFT across four hardware latency configurations, achieving ~2.8× speedup by reducing inter-stage memory traffic through efficient register-resident data reuse.

xv6 Operating System Kernel Enhancements | C, Assembly, Systems Programming | {GitHub}

- Implemented a system-call-aware Multi-Level Feedback Queue (MLFQ) scheduler in xv6, dynamically adjusting process priorities to improve responsiveness for interactive workloads while maintaining scheduling fairness and CPU efficiency.
- Implemented in memory and disk-backed swap subsystem in xv6 using RAID 0, RAID 1, and RAID 5, integrating striping, mirroring, and parity-based storage to support efficient page swapping and fault-tolerant memory management system.

Query Plan Switch Point Optimization in PostgreSQL | Python - PostgreSQL - psycopg2 - pg_hint_plan | {GitHub}

- Benchmarked PostgreSQL indexing strategies (sequential (no index), clustered, unclustered, composite) across varying selectivity, and evaluated primary key index performance based on creation timing relative to data insertion.
- Built Python script that performs linear scan across the TPC-H date range to detect PostgreSQL optimizer plan switches, then used pg_hint_plan to force each side's plan onto the other side's dates and compare runtimes, classifying switches as CORRECT, DELAYED, or EARLY; applied binary search to correct DELAYED/EARLY cases to their true switch position.

Arbitrary-Precision Arithmetic Library | *Java, Apache Ant, Docker* | *{GitHub}*

- Engineered an arbitrary-precision arithmetic library in Java supporting unlimited-size arithmetic operations.

- Designed reusable AInteger and AFloat APIs with a CLI, packaging the library as a JAR for use in Java applications.
- Automated builds with Apache Ant and containerized the application using Docker for portable deployment.

WORK EXPERIENCE

AI-Powered Data Insights Intern | TOEHO AI Technology Pvt. Ltd. | May 2026 - Aug 2026

- Architected a Python-based multi-platform ETL pipeline to collect, clean, and unify social media data from YouTube, Reddit, Twitter/X, and Instagram, processing **345K+** records across three political analytics case studies.
- Developed multilingual NLP pipelines using **XLM-RoBERTa, mDeBERTa, BERTopic, and LDA** to perform sentiment analysis, emotion classification, and topic modeling on English, Hindi, Assamese and Bengali social media data.
- Engineered graph-based analytics with **NetworkX**, computing **PageRank** and **centrality metrics** over **36K+ nodes** and **75K+ edges** to identify influential users and analyze engagement patterns.
- Optimized the analytics pipeline through **joblib-based parallel processing** and **GPU batch inference tuning**, improving large-scale processing efficiency while maintaining stable execution for **100K+** record workloads.

SCHOLASTIC ACHIEVEMENTS

- Achieved a Codeforces Specialist rating (1417) and solved 700+ coding problems across LeetCode, Codeforces, and other competitive programming platforms.
- Selected for Jawahar Navodaya Vidyalaya through JNVST, ranking among the top 80 out of 6,000+ candidates.
- Selected as a Dakshana Scholar for the 2-Year JEE Mentorship Program, among 240 out of 10,000+ students.

POSITIONS OF RESPONSIBILITY

Associate Coordinator, PR & Outreach Team | Sci-Tech Club, IIT Hyderabad Aug 2025 - Apr 2026

- Planned outreach and PR activities for club events, boosting student participation across campus.

- Collaborated with team members to execute promotional campaigns and support the successful organization of club events.

EXTRACURRICULAR

Teaching Assistant, Operating Systems-I (CS3510) & DBMS-I (CS3550) - Assist students in understanding core concepts, conduct doubt-solving sessions, provide academic guidance, and support assignments, quizzes, and evaluations.

National Sports Organization (NSO) - Athletics: Actively participated in athletics, contributing to personal development and discipline.